

- 1 (a) energy or W : $\text{kg m}^2 \text{s}^{-2}$
or
power or P : $\text{kg m}^2 \text{s}^{-3}$

M1

intensity or I : $\text{kg m}^2 \text{s}^{-2} \text{m}^{-2} \text{s}^{-1}$ (from use of energy expression)
or
 $\text{kg m}^2 \text{s}^{-3} \text{m}^{-2}$ (from use of power expression)

indication of simplification to kg s^{-3}

A1 [2]

- (b) (i) ρ : kg m^{-3} , c : m s^{-1} , f : s^{-1} , x_0 : m

M1

substitution of terms in an appropriate equation and simplification to show K has no units

A1 [2]

(ii) $I = 20 \times 1.2 \times 330 \times (260)^2 \times (0.24 \times 10^{-9})^2$

C1

$= 3.1 \times 10^{-11} (\text{W m}^{-2})$

C1

$= 31 (30.8) \text{pW m}^{-2}$

A1 [3]